

The Nagata Embedding Theorem

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Translator’s note. This is a working English translation of the uploaded LaTeX transcription of Deligne’s *Le théorème de plongement de Nagata*. It is intended as a literal mathematical translation, not a modern recast. Mathematical notation and the structure of the transcription have been kept as close as possible to the source.

Abstract. Except for a few corrections added in 2010, this article reproduces notes sent to Nagata in the 1970s, translating his papers [2] and [3] into the language of schemes. The proofs are a “constructive” version of Nagata’s proofs. Making the proofs constructive made it possible to avoid the use of valuations. A more detailed exposition and additional results may be found in Conrad [1]. Unless expressly stated otherwise, all schemes considered below are assumed to be Noetherian.

0. Recollections on blow-ups

0.1.

Let $\mathfrak{a}$ be an ideal of the ring A . The scheme obtained from $X = \mathrm{Spec}(A)$ by blowing up $\mathfrak{a}$ is the union of the open sets $D^+(x)$, where x ranges over a system of generators of $\mathfrak{a}$, and $D^+(x)$ is the spectrum of the subring $A[\mathfrak{a}x^{-1}]$ of $A[x^{-1}]$. Moreover,

$$D^+(x) \cap (X - V(\mathfrak{a})) = \mathrm{Spec}(A[x^{-1}]).$$

Let $\mathfrak{c}$ be a second ideal of A , defining a closed subscheme $V(\mathfrak{c})$ of X . The trace on $D^+(x)$ of the proper transform of $V(\mathfrak{c})$ is defined by the ideal of $A[\mathfrak{a}x^{-1}]$ obtained as the trace of the ideal of $A[x^{-1}]$ generated by $\mathfrak{c}$.

Lemma 1. *Let X be a scheme, let $\mathfrak{a}$ and $\mathfrak{b}$ be two sheaves of ideals on X , and let $\tilde{X}$ be the blow-up of X along $\mathfrak{a} + \mathfrak{b}$. Then the open subset of $\tilde{X}$ which is the union of the $D^+(x)$ for $x \in \mathfrak{b}$ contains the proper transform of $V(\mathfrak{a})$.*

Indeed, $\tilde{X}$ is the union of the $D^+(x)$ for $x \in \mathfrak{a}$ or $x \in \mathfrak{b}$, and for $x \in \mathfrak{a}$ the open set $D^+(x)$ is disjoint from the proper transform of $V(\mathfrak{a})$.

Lemma 0.3 of the original text was false. The corrected version that follows dates from March 1996. It led to a change of notation. The original lemma used the following notation, to which the rest of the text refers: instead of F and G , one considered their complements U and V , which, together with $Y' := U \cap p^{-1}(Y)$, formed a commutative diagram

$$\begin{array}{ccc} Y' & \hookrightarrow & Z \\ \downarrow & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X. \end{array}$$

Here Y' is closed in $p^{-1}(V)$.

Lemma 2. *Let $p : Z \rightarrow X$ be a morphism, let $F \supset G$ be closed subschemes of X , and let Y be a closed subscheme of Z . Assume that*

$$Y \cap p^{-1}(F) \subset p^{-1}(G) \quad \text{scheme-theoretically.}$$

Blow up X along G to obtain $\tilde{X}$, and let $\tilde{p} : \tilde{Z} \rightarrow \tilde{X}$ be obtained from $p : Z \rightarrow X$ by base change. Then the closure

$$\overline{(Y \cap p^{-1}(G) \text{ in } \tilde{Z})}$$

of $Y \cap p^{-1}(G)$ in $\tilde{Z}$ and

$$\tilde{p}^{-1}(\widetilde{F \cap G \text{ in } \tilde{X}})$$

are disjoint closed subsets of $\tilde{Z}$.

We may assume that Z and X are affine. Let G denote the ideal of G . Cover $\tilde{X}$ by the open subsets $D^+(g)$ for $g \in G$ (a system of generators of G suffices), and cover $\tilde{X} - \tilde{G}$ by the open sets

$$\tilde{U}(g) = D^+(g) \cap (\tilde{X} - \tilde{G}) \subset \tilde{X} - \tilde{G},$$

with

$$\tilde{U}(g) \subset D^+(g).$$

It is enough to see that, for each g , the traces on $\tilde{p}^{-1}D^+(g)$ of

$$\tilde{p}^{-1}(\widetilde{F \cap G \text{ in } \tilde{X}})$$

and of

$$\overline{(Y \cap p^{-1}G \text{ in } \tilde{Z})}$$

are disjoint. They are, respectively, the inverse image of the closure of $F \cap \tilde{U}(g)$ in $D^+(g)$, and the closure of $Y \cap p^{-1}(\tilde{U}(g))$ in $\tilde{p}^{-1}D^+(g)$.

Write $\sim$ for inverse image from $\tilde{X}$ to Z , or from $D^+(g)$ to $\tilde{p}^{-1}D^+(g)$. Since $p^{-1}G \supset Y \cap p^{-1}(F)$, the inverse image $\tilde{g}$ of g on $\tilde{Z}$ can be written

$$\tilde{g} = y + f,$$

where y vanishes on Y and f vanishes on $p^{-1}(F)$. There are functions a_i on $\tilde{Z}$ and elements f_i in the ideal of F such that

$$f = \sum a_i f_i \sim.$$

Since $F \supset G$, the f_i belong to the ideal of G , and the f_i/g are functions on $D^+(g)$. They vanish on $F \cap \tilde{U}(g)$, hence on

$$\overline{(F \cap \tilde{U}(g) \text{ in } D^+(g))},$$

and $\sum a_i (f_i/g) \sim$ vanishes on

$$\tilde{p}^{-1}(\widetilde{\tilde{U}(g) \text{ in } D^+(g)}).$$

On $\tilde{p}^{-1}D^+(g)$ one has

$$1 = y/g + f/g = y/g + \sum a_i (f_i/g) \sim,$$

and the same sum is therefore equal to 1 on $Y \cap p^{-1}\tilde{U}(g)$, hence on its closure

$$\overline{(Y \cap \tilde{U}(g) \text{ in } \tilde{p}^{-1}D^+(g))}.$$

This proves the required disjointness.

Remark 3. Suppose instead that we are given closed subsets $F \supset G_0$ of X , with G_0 such that $Y \cap p^{-1}(F)$ is set-theoretically contained in $p^{-1}(G_0)$. Then there exists an r such that the r -th infinitesimal neighborhood G of G_0 in F satisfies the hypotheses of the lemma. If J is the ideal defining

$$\overline{(Y \cap p^{-1}(F))}_{\text{red}}$$

in $Y \cap p^{-1}(F)$, it is enough to choose r such that $J^{r+1} = 0$.

Thus the lemma makes it possible, by a blow-up that does not affect the complement V of G_0 , to “separate” the proper transform of F and

$$\overline{(Y \cap p^{-1}(G_0))}$$

in Z .

Lemma 4. *Consider a commutative diagram of schemes*

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{p} & X \\ \uparrow & & \uparrow \\ \tilde{X}' & \xrightarrow{q} & X' \end{array}$$

with U a dense open subset of both X and X' , with p and q separated of finite type, and with $q^{-1}(X) = \tilde{X}'$.

Let F be a closed subset of X' , and let G be a closed subset of $\tilde{X}'$ containing $p^{-1}(F)$. Then there exists an ideal $\mathfrak{a}$ with

$$V(\mathfrak{a}) \subset \overline{F} - F$$

such that, after base change by the blow-up morphism defined by $\mathfrak{a}$, $\tilde{X} \rightarrow X$, and after replacing the new $\tilde{X}'$ by the closure of U , one has

$$\tilde{q}^{-1}(F) \subset \tilde{G}.$$

In Nagata’s language, to take the closure of F in the Zariski–Riemann space of X (the “bubble space”), it is enough to take the “intersection” of the closures of F in the blow-ups of X whose centers are contained in $\overline{F} - F$.

Apply Lemma 0.3 to

$$\begin{array}{ccc} \tilde{X}' & \longrightarrow & q^*F \\ \downarrow & & \\ X - F & \hookrightarrow & X - (\overline{F} - F) \hookrightarrow X. \end{array}$$

This gives $\mathfrak{a}$. After the base change $\tilde{X} \rightarrow X$, the new closure $\tilde{F}$ is disjoint from the image of the closure of the old $\tilde{X}' - q^*F$ in $\tilde{X}' - G$. In particular, after replacing $\tilde{X}'$ by $\tilde{U}$, this new closure is disjoint from the image of $\tilde{X}' - G$, which is what was required.

Lemma 5. *Let there be given a diagram of separated S -schemes of finite type*

$$\begin{array}{ccccc} & & \tilde{X}_i & & \\ & \nearrow & & \searrow & \\ U & \hookrightarrow & X_i & \longrightarrow & \tilde{X}_i \quad (1 \leq i \leq n) \\ & \searrow & & & \\ & & X & & \end{array}$$

with $\tilde{X}_i$ proper over S and U a dense open subset of X and of the $\tilde{X}_i$. Let X^* be the closure of U in the product of the $\tilde{X}_i$. Finally, let F_i be a closed subset of X_i ; assume that the intersection of the inverse images of the F_i in X^* is empty.

Then, after replacing each X_i by a blow-up with center in $\overline{F_i} - F_i$, one can obtain an analogous situation in which the intersection of the inverse images of the F_i is empty.

Let X^{**} be a blow-up of X^* ,

$$X^{**} \rightarrow \prod_i \tilde{X}_i,$$

such that in X^{**} one has $p_i^{-1}(F_i) = \emptyset$; this is obtained by blowing up $p_i^{-1}(F_i)$. Apply Lemma 0.4 to the diagrams

$$\begin{array}{ccc} p_i^{-1}(\tilde{X}_i) & \hookrightarrow & X^{**} \\ \downarrow & & \downarrow \\ U & \hookrightarrow & X_i \hookrightarrow \tilde{X}_i \end{array}$$

and to $F_i \subset X_i$, $G_i = p_i^{-1}(F_i)$. This provides the desired blow-ups of the X_i . Indeed, in the new situation, in the closure of U in X^{**} one has

$$p_i^{-1}(F_i) \subset G_i = \emptyset.$$

Since $X^{**} \rightarrow X^*$ is surjective, the same remains true in X^* .

1. The theorems

Definition 6. Let X and Y be two S -schemes, with Y separated over S . A quasi-domination $f : X \dashrightarrow Y$ is a pair consisting of a dense open subset U of X and a morphism of S -schemes $f : U \rightarrow Y$ whose graph is closed in $X \times_S Y$.

Let X and Y be two S -schemes, with Y separated over S , let U be a quasi-compact open subset of X , and let $f : U \rightarrow Y$. We say that X is quasi-dominant over Y relative to U and f if there exists a quasi-domination of the scheme-theoretic closure $\overline{U}$ of U , with values in Y , extending f . There is at most one such quasi-domination; its graph is the closure of the graph of f .

A quasi-domination f is called proper, or complete, if the morphism $f : U \rightarrow Y$ is proper.

Theorem 7 ([2, Theorem 3.2]). *Let X and Y be two S -schemes, and let $U \subset V \subset X$ be two dense open subsets of X . Assume that Y is separated of finite type over S . Let $f : U \rightarrow Y$ be a quasi-domination of V into Y . Then there exists an ideal $\mathfrak{a}$ with*

$$V(\mathfrak{a}) \subset X - V$$

such that the blow-up $\tilde{X}$ of X along $V(\mathfrak{a})$ quasi-dominates Y .

By the base change $X \rightarrow S$, one reduces to the case $X = S$. Then Y is a scheme over X , f is a section of Y over U , and by hypothesis $f(U)$ is closed over V .

Case 1. $U = V$, $p : Y \rightarrow X$ quasi-affine, X affine.

By hypothesis there is a commutative diagram

$$\begin{array}{ccc} Y & \xhookrightarrow{\quad} & E \\ & \searrow p & \swarrow \\ & X & \end{array}$$

with E an affine space of finite type over X . If X quasi-dominates E relative to f , then *a fortiori* X quasi-dominates Y . Thus we may assume $Y = E_X$. A preliminary blow-up makes it possible to assume that $X - U$ is a divisor D . Locally on X , the section f therefore has coordinates $(f_i u^{-n})$, where u is a local equation for D , the f_i are regular, and n is chosen large enough once and for all. Let $\mathfrak{a}$ be the ideal generated by u and the f_i . After blowing up $V(\mathfrak{a})$, the section f extends to a section of the projective completion of E_X , and this proves the case.

Case 2. $U = V$, p quasi-affine.

By Case 1, if $(U_i)_{0 \leq i \leq n}$ is an affine open covering of X , then on each U_i there exists an ideal $\mathfrak{a}_i$ which works. This ideal has an extension $\tilde{\mathfrak{a}}_i$ to X , with $V(\tilde{\mathfrak{a}}_i) \subset X - V$, and it is enough to blow up the product of the $\tilde{\mathfrak{a}}_i$.

Case 3. p quasi-affine.

By Lemma 0.3, applied to

$$\begin{array}{ccc} f(U) & \hookrightarrow & Y \\ \downarrow & & \downarrow p \\ U & \hookrightarrow & V \hookrightarrow X, \end{array}$$

a preliminary blow-up makes it possible to assume that

$$p(\overline{f(U)}) \cap (V - U) = \emptyset.$$

There is therefore a quasi-compact open subset X' of X with $X' \cap V = U$ and $p(\overline{f(U)}) \subset X'$. By Case 2, there exists an ideal $\mathfrak{a}$ on X' , with $V(\mathfrak{a}) \subset X' - U$, such that the blow-up $\widetilde{X'}$ quasi-dominates Y , equivalently $\overline{f(U)}$. It remains only to extend $\mathfrak{a}$ to an ideal on X , with $V(\mathfrak{a}) \subset X - V$.

General case.

Let (Y_i) be a finite covering of Y by affine open subsets, and set $U_i = f^{-1}(Y_i)$, $X_i = \overline{U_i}$, $V_i = X_i \cap V$.

Case 3 applies to the diagrams

$$\begin{array}{ccccc} U_i & \hookrightarrow & V_i & \hookrightarrow & X_i \\ f_i \downarrow & & & & \downarrow \\ Y_i & \hookrightarrow & Y & & X, \end{array}$$

and gives an ideal $\mathfrak{a}_i$ on X_i . This ideal defines an ideal, still denoted $\mathfrak{a}_i$, on X , again with $V(\mathfrak{a}_i) \subset X - V$. Blow up the product of the $\mathfrak{a}_i$. The resulting blow-up $q : \tilde{X} \rightarrow X$ is such that $q^{-1}(X_i)$ dominates the blow-up $\tilde{X}_i$. If $X'_i = q^{-1}(X_i)$, there exists an open subset U'_i of X'_i , with $U'_i \supset q^{-1}(U_i)$, such that, over X'_i , $f(U'_i)$ is a section of $f : U'_i \rightarrow Y$. We conclude that after the blow-up,

$$f(U) = \bigcup f_i(U'_i).$$

To prove Theorem 1.2, we may replace Y by $\overline{f(U)}$, and we conclude by Case 3 and the following lemma.

Lemma 8. $\overline{f(U)}$ is quasi-affine over X .

Indeed, $\overline{f(U)}$ is quasi-finite over X .

Corollary 9 (A variant of Chow's lemma). *Let $f : X \rightarrow S$ be a separated morphism of finite type. Then for every open subset U which is quasi-projective over S and dense in X , there exists a diagram of S -schemes*

$$X \xleftarrow{q} \tilde{X}' \xrightarrow{j} \tilde{X}$$

where q is a blow-up morphism defined by an ideal satisfying $V(\mathfrak{a}) \subset \tilde{X}' - U$, and where $\tilde{X}$ is projective over S .

In particular, q is projective and surjective, and $q^{-1}(U) \xrightarrow{\sim} U$.

Let U^* be a projective S -scheme containing U as a schematically dense open subset. By Theorem 1.2 applied to the inclusion of U in X , one can choose U^* quasi-dominant over X : one has a diagram

$$\begin{array}{ccc} & U & \\ \swarrow & & \searrow \\ V & \xleftarrow{\phi} & U^* \\ \searrow & & \swarrow \\ & \bar{U} & \longrightarrow X, \end{array} \quad (1)$$

and here, since U^* is proper over S , ϕ is proper.

Lemma 10. *Let there be given a diagram of separated S -schemes of finite type*

$$\begin{array}{ccc} U & \hookrightarrow & V \\ \downarrow & & \downarrow \\ X & \xleftarrow{\phi} & Y \end{array}$$

with U dense in X , dense in Y , and schematically dense in V , and with ϕ a quasi-domination of Y into X . Let $\mathfrak{a}$ be an ideal of X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\tilde{X}$ of X along $\mathfrak{a}$ quasi-dominates V relative to U , and such that U is schematically dense in $\tilde{X}$. Thus we obtain a diagram

$$\begin{array}{ccc} & \tilde{U} & \\ \swarrow & & \searrow \\ \tilde{V} & \xleftarrow{\beta} & \tilde{W} \\ \searrow \psi & & \swarrow \alpha \\ & \tilde{X}. & \end{array}$$

Then:

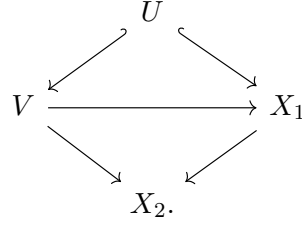
- (a) ψ is the blow-up morphism of V along $\phi^*\mathfrak{a}$;
- (b) the graph of ψ is closed in $\tilde{X} \times_S Y$.

Let $\tilde{V}$ be the blow-up of V along $\phi^*\mathfrak{a}$.

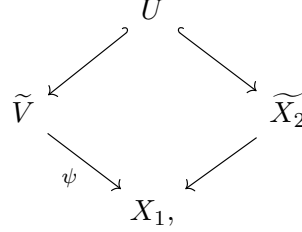
- (1) By the universal property of $\tilde{X}$, there is a unique $\tilde{X}$ -morphism from $\tilde{V}$ to $\tilde{X}$; denote it by α .
- (2) By the universal property of $\tilde{V}$, there is a unique V -morphism from $\tilde{W}$ to $\tilde{V}$; denote it by β .

$$\begin{array}{ccc} \tilde{W} & \xrightarrow{\beta} & \tilde{V} \\ \alpha \downarrow & & \downarrow \psi \\ \tilde{X} & & V \hookrightarrow Y \end{array} \quad \begin{array}{ccc} U & \hookrightarrow & \tilde{W} \\ \downarrow & & \downarrow \\ V & \longrightarrow & \tilde{V} \longrightarrow \tilde{X}. \end{array}$$

By Theorem 1.2, after blowing up X_1 without changing U , we reduce to the case where X_1 quasi-dominates X_2 .



Apply Lemma 1.5 to this diagram so as to obtain



where

- (a) ψ is the blow-up morphism of an ideal $\mathfrak{a}$ of V with $V(\mathfrak{a}) \subset V - U$;
- (b) the graph of ψ is closed in $X_1 \times_S \tilde{X}_2$; and
- (c) $\tilde{X}_2$ is a blow-up of X_2 .

Let $\mathfrak{a}'$ be an ideal of X_1 extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset X_1 - U$, and let $q : \tilde{X}_1' \rightarrow X_1$ be the blow-up of X_1 along $\mathfrak{a}'$. Then ψ induces an isomorphism between $\tilde{W}$ and $q^{-1}(V)$, and the graph of this isomorphism is closed in $\tilde{X}_1' \times \tilde{X}_2$. Define $\tilde{X}$ by gluing $\tilde{X}_1'$ and $\tilde{X}_2$ along this isomorphism.

Remark 11. The proof shows that, for the maps $p_i : U_i \rightarrow X_i$ with $U_i \subset X_i$, one may take a blow-up morphism of an ideal supported outside U .

Theorem 12 ([2, Lemma 4.1 and Theorem 4.3]). *Let $f : X \rightarrow S$ be separated of finite type. Then there exists an immersion of X into an S -scheme proper over S .*

The graph of ϕ is, trivially, proper over V ; the same is true for its inverse image on $V \times_S X$, and it follows that the graph of ψ is proper over V , hence that ψ is proper. Thus β is proper, and has schematically dense image, since U is already schematically dense in V . On the other hand, β is a local section of α on $\tilde{W}$ because $\beta|_U$ is one on U ; hence β is an isomorphism.

By hypothesis, the graph of ψ is closed in $V \times_S X$, and is contained in the inverse image of the graph of ϕ , which is closed in $Y \times X$. It follows that $\Gamma(\psi)$ is closed in $Y \times \tilde{X}$.

Apply Lemma 1.5 to (1). There exists an ideal $\mathfrak{a}$ on X , with $V(\mathfrak{a}) \subset X - U$, such that the blow-up $\tilde{X}$ quasi-dominates $\bar{U} \subset X - V$. Moreover, one can arrange that U is schematically dense in $\tilde{X}$. By Lemma 1.5, and because ϕ is proper, $\tilde{X}$ is therefore the blow-up of V along $\phi^*\mathfrak{a}$. If $\mathfrak{a}'$ is an ideal of U^* extending $\mathfrak{a}$, with $V(\mathfrak{a}') \subset U^* - U$, we then take

$$\tilde{X}' = \tilde{X}, \quad \tilde{X} = U^*.$$

Lemma 13 ([2, Lemma 4.2]). *Let there be given a diagram of separated S -schemes of finite type*

$$X_2 \longleftarrow U \longrightarrow X_1$$

with U a dense open subset of the X_i . Then there exists an embedding $j : U \hookrightarrow X$, with X separated of finite type over S and U schematically dense in X , and there exist proper quasi-dominations $p_i : X \dashrightarrow X_i$.

Let U_i be a finite open covering of X by dense quasi-projective open subsets. For each U_i there exists, by Corollary 1.4, a commutative diagram of S -schemes

$$\begin{array}{ccc} & & \tilde{X}_i \\ & \nearrow j_i & \downarrow \\ U_i & \xrightarrow{q_i} & X_i \end{array}$$

with q_i proper, $\tilde{X}_i$ proper over S , and U_i dense in $\tilde{X}_i$. Let $F_i = X_i - U_i$ and $F'_i = q_i^{-1}(F_i)$. Finally, let X^* be the closure of U in the product of the $\tilde{X}_i$, and write $p_i : X^* \rightarrow \tilde{X}_i$.

By Lemma 0.5, after a preliminary blow-up of X_i which does not affect $\tilde{X}'_i$, we may assume that

$$p_i^{-1}(F'_i) = \emptyset.$$

Let M be the separated S -scheme obtained by gluing the $\tilde{X}_i$ and $X_i - F'_i$ along U_i .

$$\begin{array}{ccc} X_i - F'_i & & \\ \downarrow & & \\ \tilde{X}_i & \longrightarrow & M. \end{array}$$

Apply Lemma 1.7 to X and to the M_i , to obtain $X \rightarrow M$ where M properly quasi-dominates each M_i . It remains to prove that M is proper over S .

Indeed, one has maps

$$X^* - p_i^{-1}(F'_i) \rightarrow M_i$$

which glue over $U = \bigcup U_i$. After replacing X^* by a blow-up X^{**} , one obtains maps

$$X^{**} - p_i^{-1}(F'_i) \rightarrow M_i$$

which glue. This gives a map from the scheme-theoretic closure of U in X^{**} with values in M . The scheme M , being the image of this map, is proper over S .

Comments

I hope that, by being still more constructive, one can prove Theorem 1.6 for $f : X \rightarrow S$ separated of finite type with S quasi-compact and quasi-separated. The rules to follow are:

- (a) consider only quasi-compact open subsets and closed subsets defined by ideals of finite type;
- (b) never take a scheme-theoretic closure, since that would take us outside (a); instead, take only a closed subset containing the given part and sufficiently small for the later constructions;
- (c) do not speak of a dense quasi-projective open subset; cf. (b). Remember that, to construct such open subsets, one could start with an affine cover U_i and take $U'_i = U_i - \bigcup_{j < i} U_j$. Translate the arguments in terms of the original U_i , and apply (a)+(b).

On the other hand, I have no idea in the case of algebraic spaces.

Acknowledgments

The original notes were handwritten. They were typed by V. Maillot.

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